

Runs scored in	Over	1	2	3	4	5	6
every over	Runs	7	3	0	9	6	10
					↓	↓	↓
					19	25	35

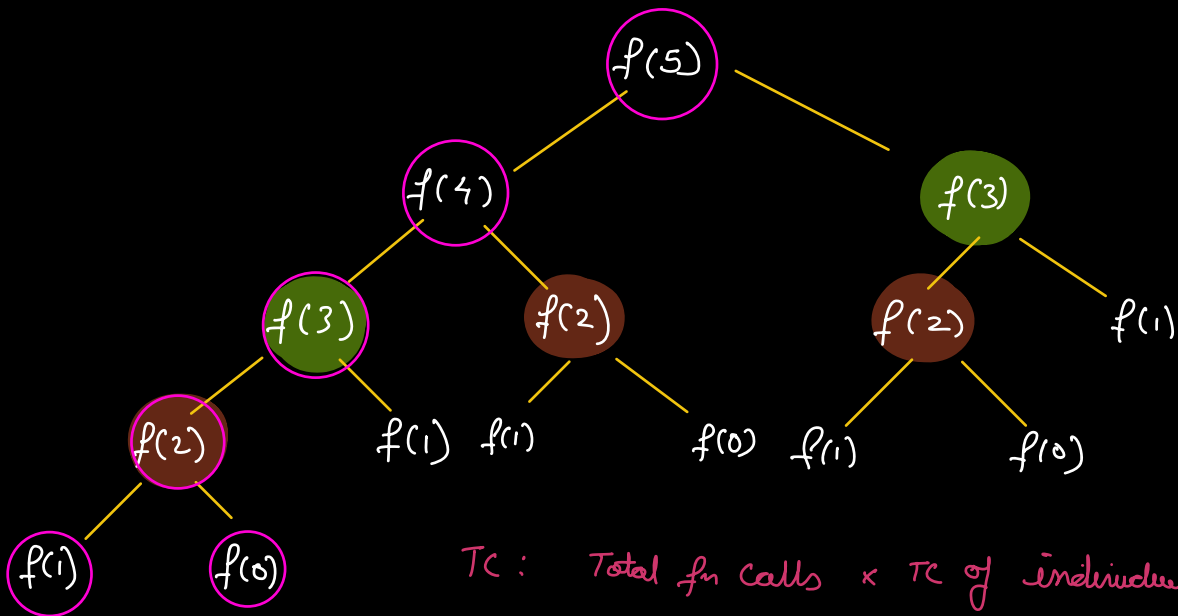
Those who do not remember the past are considered to repeat it.

fib : 0 1 1 2 3 5 8 13 ...

```

int fib(N) {
    if (N <= 1) {
        return N;
    }
    return fib(N-1) + fib(N-2);
}

```



TC: Total fn calls $\times$ TC of individual fn call
 $2^n \times O(1)$
 $\Rightarrow O(2^n)$ SC: $O(N)$

```
int F[N+1] = {-1}
```

```
int fib(N) {
```

```
// Not calculated yet
```

```
if (F[N] == -1) {
```

```
    if (N <= 1)
```

```
        F[N] = N;
```

```
    else
```

```
        F[N] = fib(N-1) + fib(N-2);
```

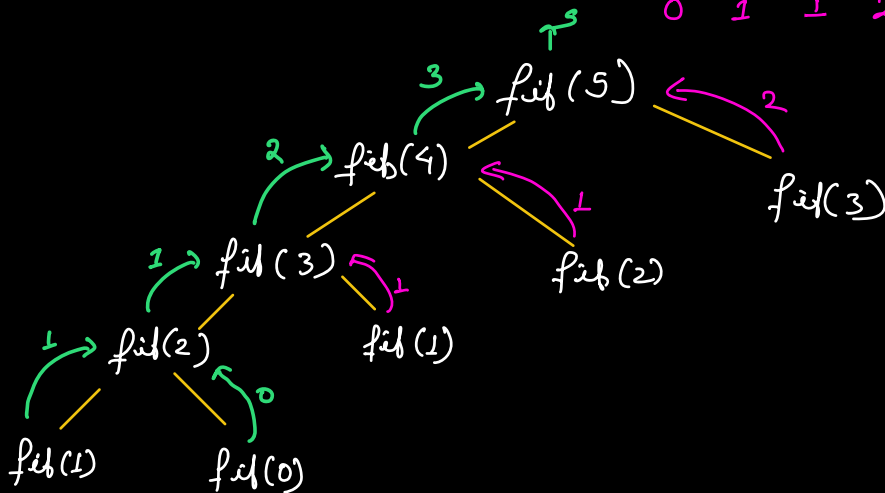
```
}
```

```
return F[N];
```

```
}
```

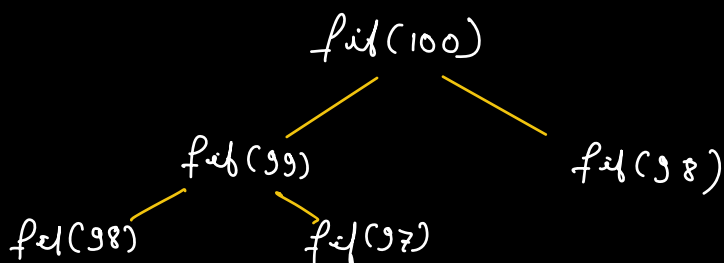
F :

0	1	2	3	4	5
-1	-1	-1	-1	-1	-1
0	1	1	2	3	5



TC : $O(N)$
SC : $O(N)$

↓
Top-down DP
DP with memoization



	0	1	2	3	4	5
F;	0	1	1	2	3	5

```

fib(N) {
    int F[N+1];
    F[0] = 0; F[1] = 1;
    for (i=2; i<=N; i++){
        F[i] = F[i-1] + F[i-2];
    }
    return F[N];
}

```

{ Bottom Up
 Tabulation
 Iterative DP

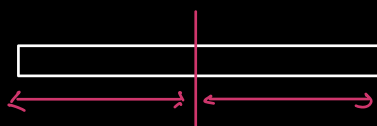
```

fib(N) {
    int a=0, b=1, c;
    for (i=2; i<=N; i++){
        c = a+b;
        a = b;
        b = c;
    }
    return c;
}

```

a	b	c
0	1	1
1	1	2
1	2	3
2	3	5

Recursion



Sort(A)

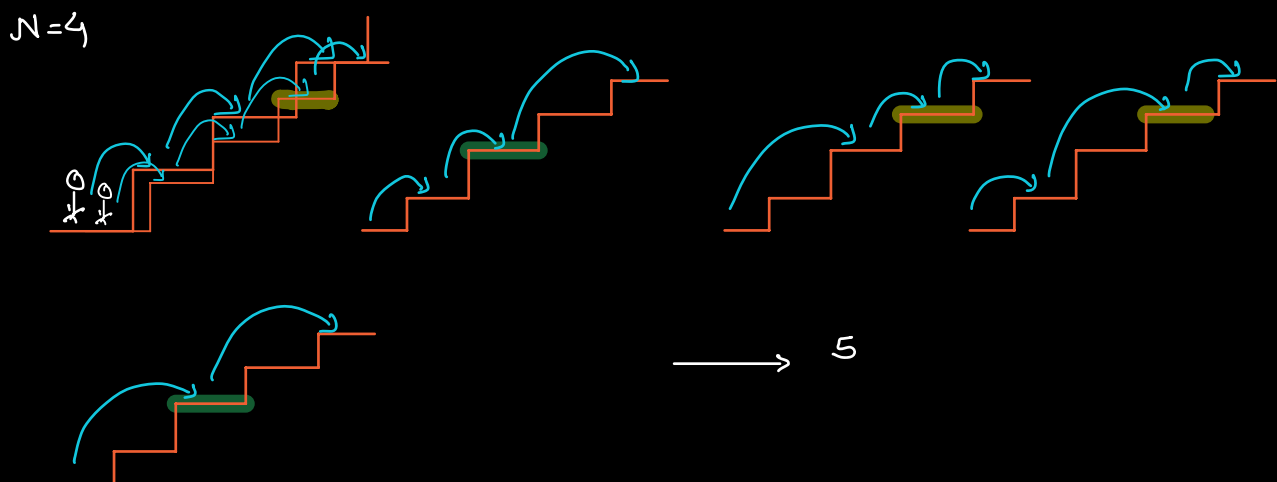
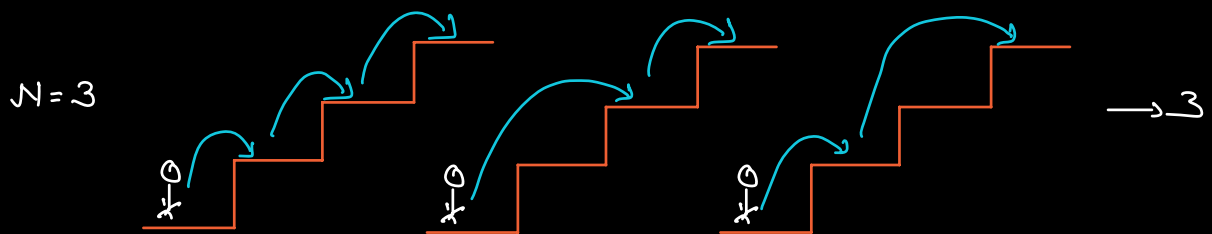
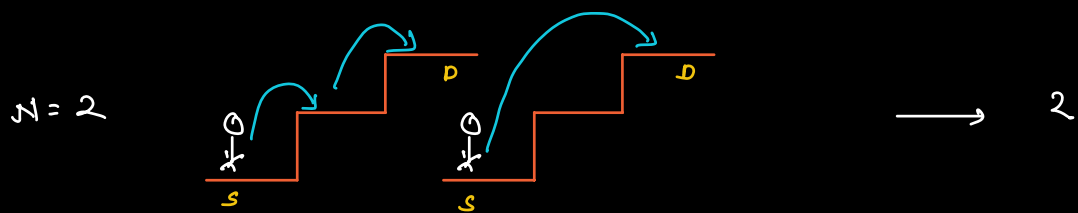
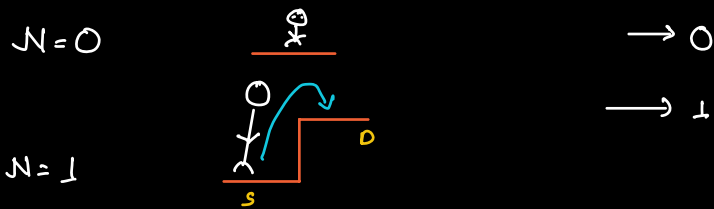
Sort (first half)

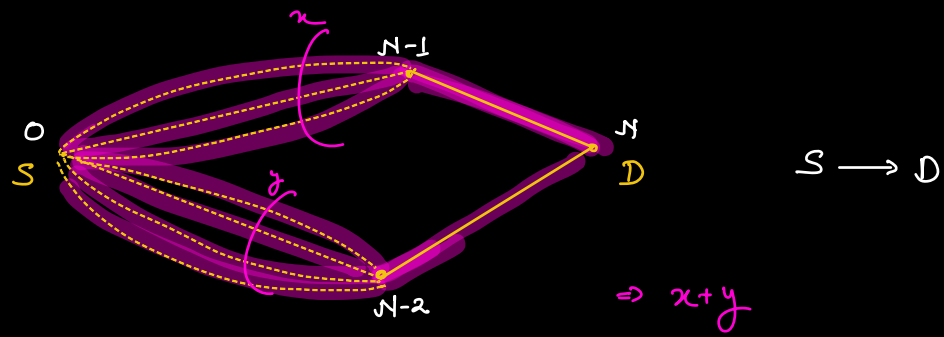
Sort (2nd half)

merge (2 half)

Amazon, MS, Adobe, Flipkart, Samsung, Zeta/Medias.net / Direct i
Google, Uber, Ola, Myntra - - - .

Q 1 Given N stairs. From any i^{th} step you can either go
($i+1$)th step or ($i+2$)th step.
Count the no of ways of going from 0th to N th step.

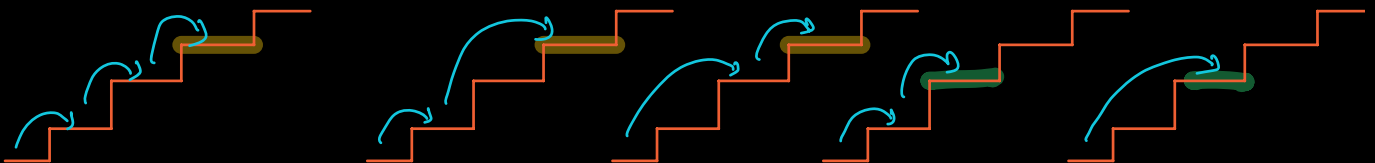
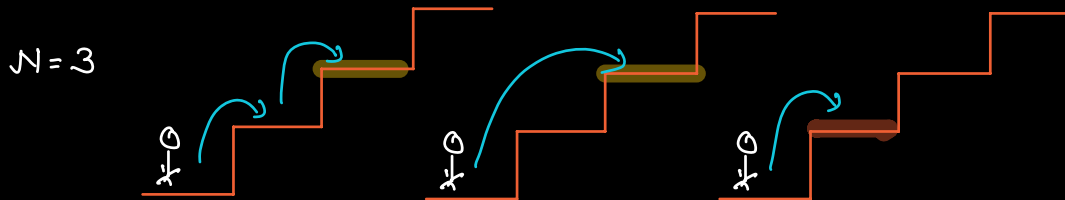


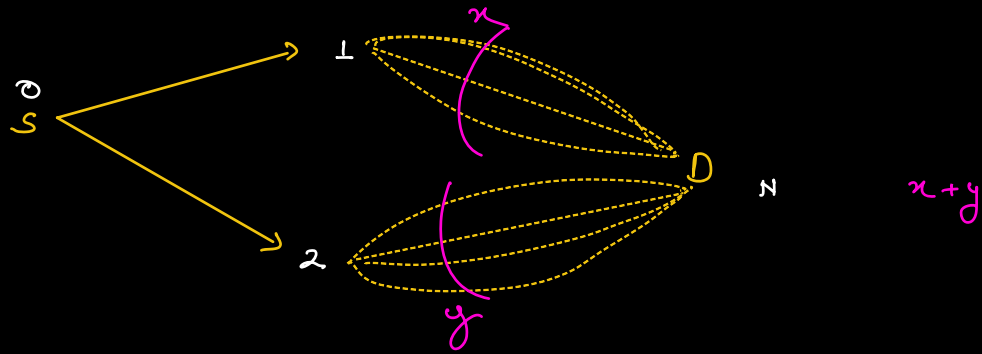


$$\text{ways}[0 \rightarrow N] = \text{ways}[0 \rightarrow N-1] + \text{ways}[0 \rightarrow N-2]$$

$\text{ways}(i) \rightarrow$ no. of ways to reach i from 0

$$\text{ways}(N) = \text{ways}(N-1) + \text{ways}(N-2)$$





$$\text{ways}[0 \rightarrow N] = \text{ways}[1 \rightarrow N] + \text{ways}[2 \rightarrow N]$$

$\text{ways}(i) \longrightarrow$ No. of ways to reach N^{th} step from i^{th} step

$$\text{ways}(0) = \text{ways}(1) + \text{ways}(2)$$

HW : Dry Run Memoization

Q House Robber



Given an array (+ve no). Return the max sum without any adjacent element.

Approach 1

Wrong!!

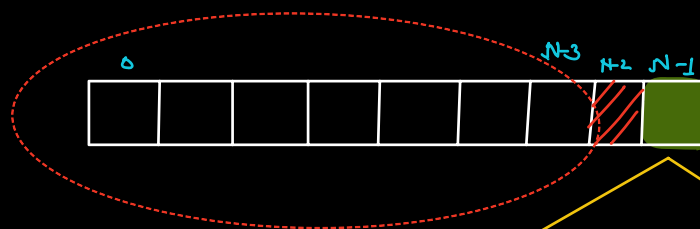
$$\text{Ans} = \max \left(\begin{array}{l} \text{Sum of all even indexed elements} \\ \text{Sum of all odd indexed elements} \end{array} \right)$$

0	1	2	3	4	5
1	1	100	2	1	100

$$\Sigma \text{ even indexed elements} = 102$$

$$\Sigma \text{ odd indexed elements} = 103$$

$$\text{Ans} \longrightarrow 201$$

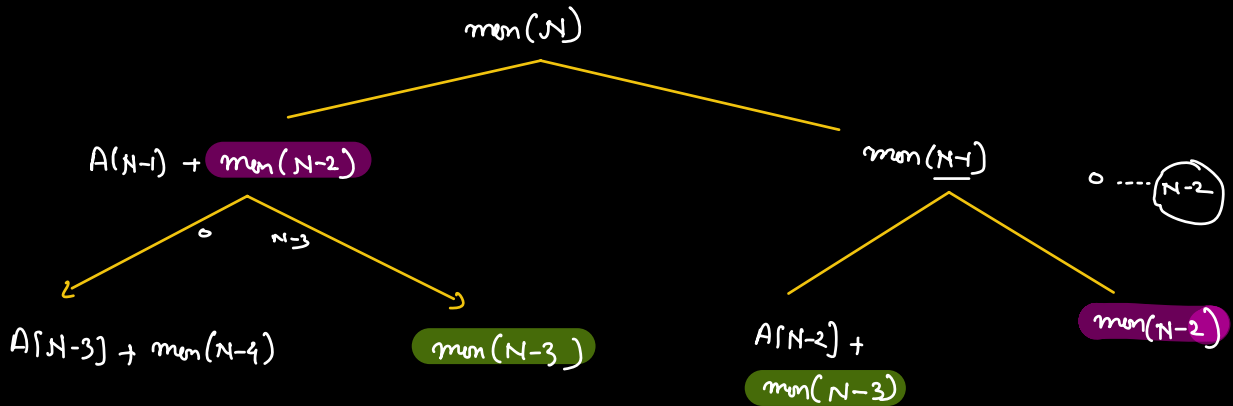


$$\begin{array}{c} \text{Current} \\ \downarrow \\ \text{money}(N) \\ [0, N-1] \end{array}$$

$$\text{Ans} = \max \left(\begin{array}{l} A[N-1] + \text{money}(N-2) \\ \text{money}(N-1) \end{array} \right)$$

$\text{money}(i) \longrightarrow$ Max money that can be robbed from first i house (index $\rightarrow [0, i-1]$)

$$\text{money}(N) = \max(A(N-1) + \text{money}(N-2), \text{money}(N-1)),$$

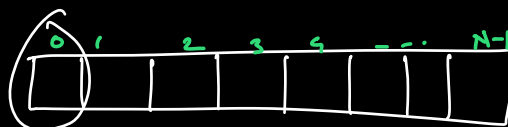
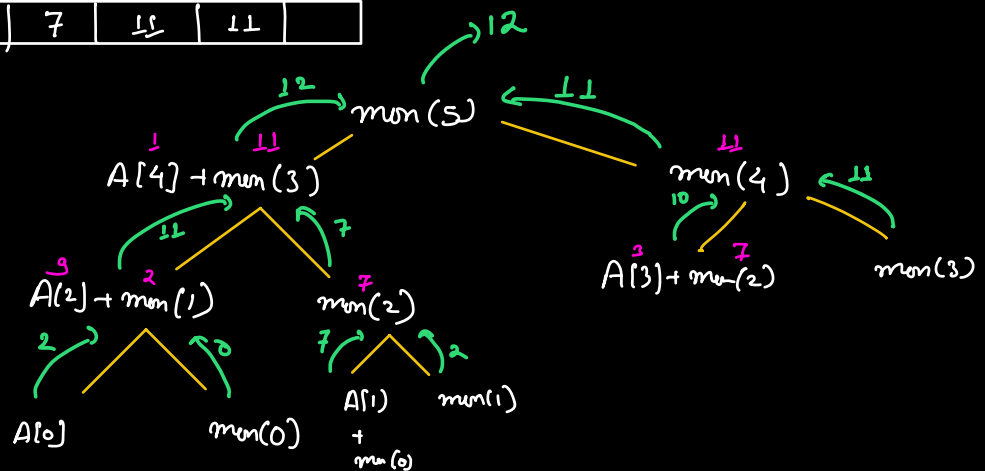


Overlapping
Sub-problems

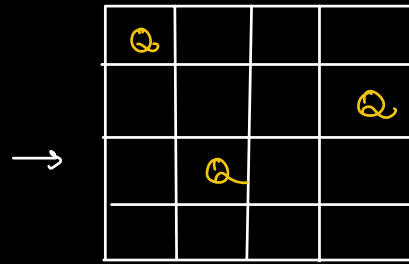
Ex

0	1	2	3	4
2	7	9	3	1

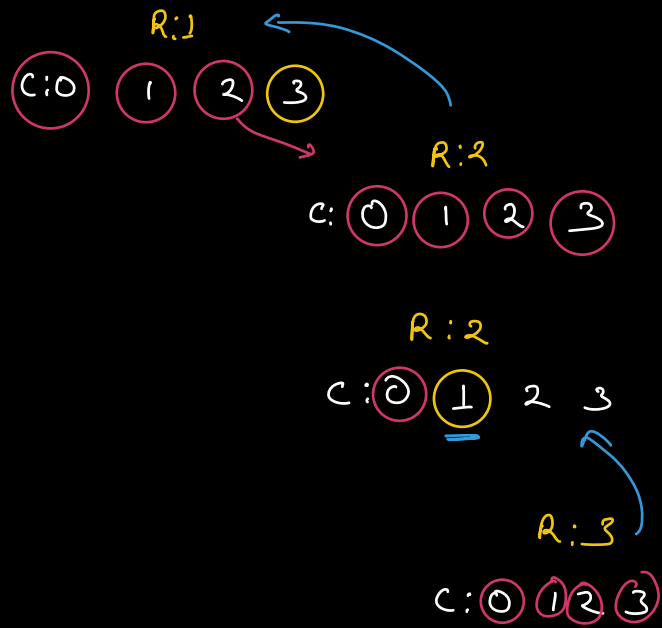
0	1	2	3	4	5
0	2	7	15	11	



$$\text{money}(0 \rightarrow N-1) = A[0] + \text{money}(2 \rightarrow N-1), \quad \text{money}(1 \rightarrow N-1)$$



$R:0$
 $C:0$ $C:1$ $C:2$ $C:3$



```

boolean NQueen (mat, N, row) {
    if (row == N) {
        return true;
    }
    for (col = 0; col < N; col++) {
        if (isSafe(mat, row, col)) {
            // Place the queen at row, col
            mat[row][col] = 1;
            if (NQueen(mat, N, row+1)) {
                return true;
            }
            mat[row][col] = 0;
        }
    }
    return false;
}

```

R:0 →

	0	1	2	3
0	1	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0

← false

R: →

	0	1	2	3
0	1	0	0	0
1	0	0	1	0
2	0	0	0	0
3	0	0	0	0

← false

	0	1	2	3
0	1	0	0	0
1	0	0	0	1
2	0	0	0	0
3	0	0	0	0

R: $\rightarrow$

	0	1	2	3
0	1	0	0	0
1	0	0	1	0
2	0	0	0	0
3	0	0	0	0

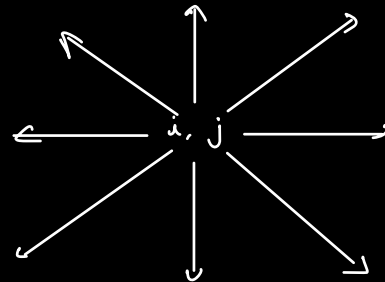
False
R: $\rightarrow$

	0	1	2	3
0	1	0	0	0
1	0	0	0	1
2	0	1	0	0
3	0	0	0	0

R: $\rightarrow$

	0	1	2	3
0	1	0	0	0
1	0	0	0	1
2	0	1	0	0
3	0	0	0	0

	0	1	2	3
0	1	0	0	0
1	0	0	1	0
2	0	0	0	0
3	0	0	0	0



	0	1	2	3	4	5	6
0	00	01	02	03	04	05	06
1	10	11	12	13	14	15	16
2	20	21	22	23	24	25	26
3	30	31	32	33	34	35	36
4	40	41	42	43	44	45	46
5	50	51	52	53	54	55	56
6	60	61	62	63	64	65	66

TC: $O(N)$

Col $\rightarrow O(1)$

$N \times N$

$i+j \rightarrow$ Constant for a given dia
& unique for every dia

	0	1	2	3	4	5	6
0	00	01	02	03	04	05	06
1	10	11	12	13	14	15	16
2	20	21	22	23	24	25	26
3	30	31	32	33	34	35	36
4	40	41	42	43	44	45	46
5	50	51	52	53	54	55	56
6	60	61	62	63	64	65	66

N x N

$i - j \rightarrow$ constant for a given diag
 unique for every diag.

colSet $\rightarrow j \quad \{2\}$
 diag1 set $\rightarrow i + j \quad \{6\}$
 diag2 set $\rightarrow i - j \quad \{2\}$